Chapter 9
Three-Level and Mixed-Level Factorial and Fractional Factorial Design Solutions

9.1S. The effects of developer strength (*A*) and developer time (*B*) on the density of photographic plate film are being studied. Three strengths and three times are used, and four replicates of a 3^2 factorial experiment are run. The data from this experiment follow. Analyze the data using the standard methods for factorial experiments.

	Development Time (minutes)					
Developer Strength	10		14		18	
1	0	2	1	3	2	5
	5	4	4	2	4	6
2	4	6	6	8	9	10
	7	5	7	7	8	5
3	7	10	10	10	12	10
	8	7	8	7	9	8

Design Expert Output

Response: Data

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	224.22	8	28.03	10.66	< 0.0001	significant
A	*198.22*	*2*	*99.11*	*37.69*	*< 0.0001*	
B	*22.72*	*2*	*11.36*	*4.32*	*0.0236*	
AB	*3.28*	*4*	*0.82*	*0.31*	*0.8677*	
Pure Error	71.00	27	2.63			
Cor Total	295.22	35				

The Model F-value of 10.66 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B are significant model terms.

Strength and time are significant. The quadratic and interaction effects are not significant. By treating both *A* and *B* as numerical factors, the analysis can be performed as follows:

Design Expert Output

Response: Data

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	214.71	2	107.35	44.00	< 0.0001	significant
A	*192.67*	*1*	*192.67*	*78.97*	*< 0.0001*	
B	*22.04*	*1*	*22.04*	*9.03*	*0.0050*	
Residual	80.51	33	2.44			
Lack of Fit	*9.51*	*6*	*1.59*	*0.60*	*0.7255*	*not significant*
Pure Error	*71.00*	*27*	*2.63*			
Cor Total	295.22	35				

The Model F-value of 44.00 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

9.2. Compute the *I* and *J* components of the two-factor interaction in Problem 9.1.

	B		
A	11	10	17
	22	28	32
	32	35	39

$$AB \text{ Totals} = 77,\ 78,\ 71;\ I(AB) = \frac{77^2 + 78^2 + 71^2}{12} - \frac{226^2}{36} = 2.39$$

$$AB^2 \text{ Totals} = 78,\ 74,\ 74;\ J(AB) = \frac{78^2 + 74^2 + 74^2}{12} - \frac{226^2}{36} = 0.89$$

$$SS_{AB} = I(AB) + J(AB) = 3.28$$

9.3S. An experiment was performed to study the effect of three different types of 32-ounce bottles (*A*) and three different shelf types (*B*) -- smooth permanent shelves, end-aisle displays with grilled shelves, and beverage coolers -- on the time it takes to stock ten 12-bottle cases on the shelves. Three workers (factor *C*) were employed in this experiment, and two replicates of a 3^3 factorial design were run. The observed time data are shown in the following table. Analyze the data and draw conclusions.

		Replicate I			Replicate II		
Worker	Bottle Type	Permanent	End Aisle	Cooler	Permanent	End Aisle	Cooler
1	Plastic	3.45	4.14	5.80	3.36	4.19	5.23
	28-mm glass	4.07	4.38	5.48	3.52	4.26	4.85
	38-mm glass	4.20	4.26	5.67	3.68	4.37	5.58
2	Plastic	4.80	5.22	6.21	4.40	4.70	5.88
	28-mm glass	4.52	5.15	6.25	4.44	4.65	6.20
	38-mm glass	4.96	5.17	6.03	4.39	4.75	6.38
3	Plastic	4.08	3.94	5.14	3.65	4.08	4.49
	28-mm glass	4.30	4.53	4.99	4.04	4.08	4.59
	38-mm glass	4.17	4.86	4.85	3.88	4.48	4.90

Design Expert Output

Response: Time

ANOVA for Selected Factorial Model

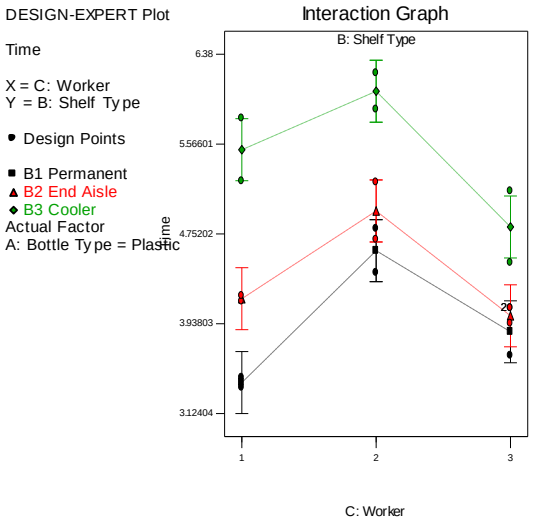
Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Model	28.28	26	1.09	14.50	< 0.0001	significant
A	*0.41*	*2*	*0.21*	*2.74*	*0.0828*	
B	*17.75*	*2*	*8.88*	*118.34*	*< 0.0001*	
C	*7.66*	*2*	*3.83*	*51.09*	*< 0.0001*	
AB	*0.12*	*4*	*0.029*	*0.39*	*0.8163*	
AC	*0.11*	*4*	*0.027*	*0.36*	*0.8319*	
BC	*1.68*	*4*	*0.42*	*5.60*	*0.0021*	
ABC	*0.55*	*8*	*0.069*	*0.92*	*0.5145*	
Pure Error	2.03	27	0.075			
Cor Total	30.31	53				

The Model F-value of 14.50 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case B, C, BC are significant model terms.

Factors *B* and *C*, shelf type and worker, and the *BC* interaction are significant. For the shortest time regardless of worker chose the permanent shelves. This can easily be seen in the interaction plot below.



9.4. An experiment is run in a chemical process using a 3^2 factorial design. The design factors are temperature and pressure, and the response variable is yield. The data that result from this experiment are shown below.

	Pressure, psig		
Temperature, °C	100	120	140
80	47.58, 48.77	64.97, 69.22	80.92, 72.60
90	51.86, 82.43	88.47, 84.23	93.95, 88.54
100	71.18, 92.77	96.57, 88.72	76.58, 83.04

(a) Analyze the data from this experiment by conducting an analysis of variance. What conclusions can you draw?

Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	3187.13	8	398.39	4.37	0.0205	significant
A	*1096.93*	*2*	*548.47*	*6.02*	*0.0219*	
B	*1503.56*	*2*	*751.78*	*8.25*	*0.0092*	
AB	*586.64*	*4*	*146.66*	*1.61*	*0.2536*	
Pure Error	819.98	9	91.11			
Cor Total	4007.10	17				

The Model F-value of 4.37 implies the model is significant. There is only a 2.05% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B are significant model terms.

Temperature and pressure are significant. Their interaction is not. An alternate analysis is performed below with *A* and *B* treated as numeric factors:

Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

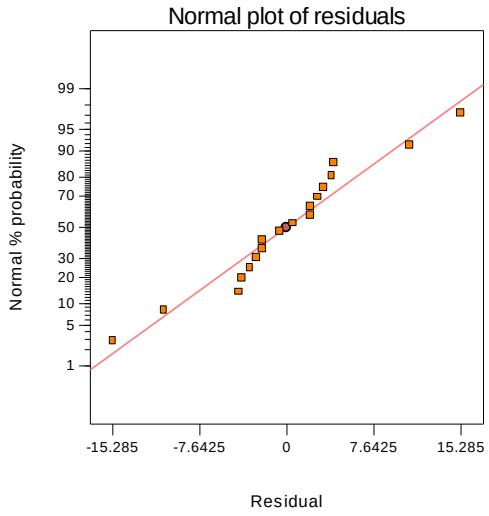
Source	Sum of Squares	DF	Mean Square	F Value	Prob > F
Model	3073.27	5	614.65	7.90	0.0017 significant
A	*850.76*	*1*	*850.76*	*10.93*	*0.0063*
B	*1297.92*	*1*	*1297.92*	*16.68*	*0.0015*
A2	*246.18*	*1*	*246.18*	*3.16*	*0.1006*
B2	*205.64*	*1*	*205.64*	*2.64*	*0.1300*
AB	*472.78*	*1*	*472.78*	*6.08*	*0.0298*
Residual	933.83	12	77.82		
Lack of Fit	*113.86*	*3*	*37.95*	*0.42*	*0.7454 not significant*
Pure Error	*819.98*	*9*	*91.11*		
Cor Total	4007.10	17			

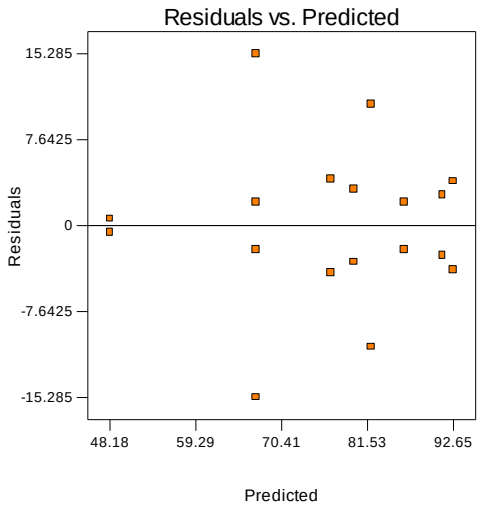
The Model F-value of 7.90 implies the model is significant. There is only a 0.17% chance that a "Model F-Value" this large could occur due to noise.

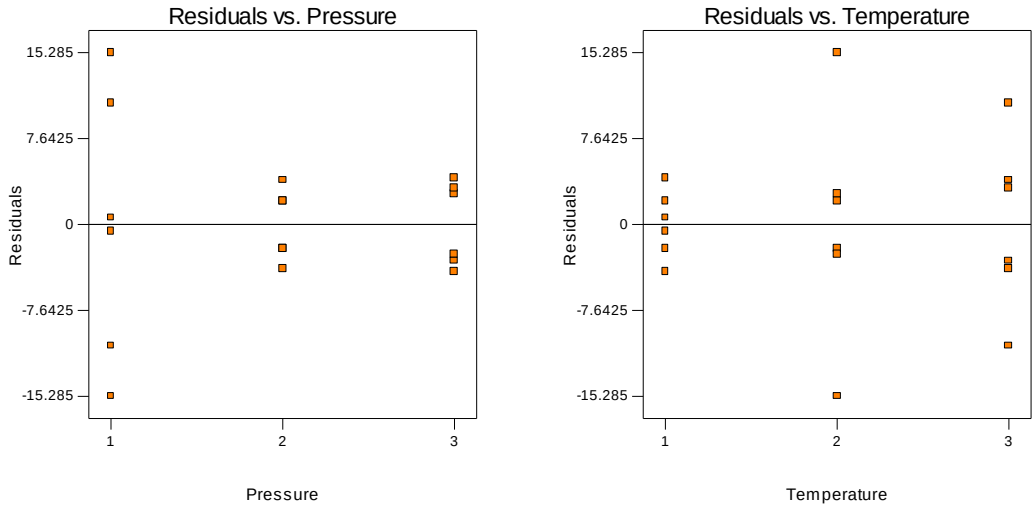
Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B, AB are significant model terms.

(b) Graphically analyze the residuals. Are there any concerns about underlying assumptions or model adequacy?

The following residual plots are based on the first analysis shown above with the *A* and *B* treated as categorical factors. The plot of residuals versus pressure shows a decreasing funnel shape indicating a non-constant variance.







(c) Verify that if we let the low, medium and high levels of both factors in this experiment take on the levels -1, 0, and +1, then a least squares fit to a second order model for yield is

$$\hat{y} = 86.81 + 10.4x_1 + 8.42x_2 - 7.17x_1^2 - 7.84x_2^2 - 7.69x_1x_2$$

The coefficients can be found in the following table of computer output.

Design Expert Output

Final Equation in Terms of Coded Factors:

Yield	=	
+86.81		
+8.42	* A	
+10.40	* B	
-7.84	* A^2	
-7.17	* B^2	
-7.69	* A * B	

(d) Confirm that the model in part (c) can be written in terms of the natural variables temperature (T) and pressure (P) as

$$\hat{y} = -1335.63 + 18.56T + 8.59P - 0.072T^2 - 0.0196P^2 - 0.0384TP$$

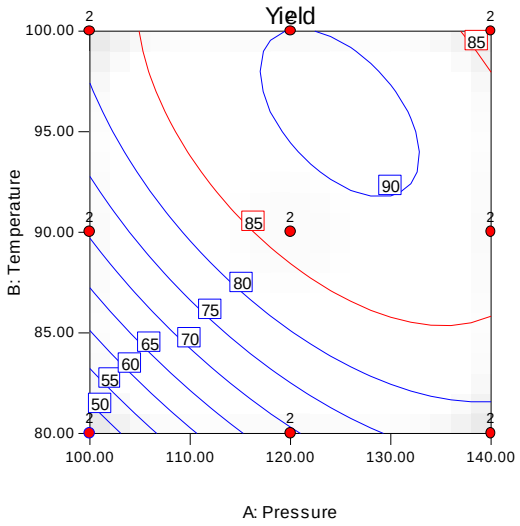
The coefficients can be found in the following table of computer output.

Design Expert Output

Final Equation in Terms of Actual Factors:

Yield	=	
-1335.62500		
+8.58737	* Pressure	
+18.55850	* Temperature	
-0.019612	* Pressure^2	
-0.071700	* Temperature^2	
-0.038437	* Pressure * Temperature	

(e) Construct a contour plot for yield as a function of pressure and temperature. Based on the examination of this plot, where would you recommend running the process?



Run the process in the oval region indicated by the yield of 90.

9.5.

(a) Confound a 3^3 design in three blocks using the ABC^2 component of the three-factor interaction. Compare your results with the design in Figure 9.7.

$$L = x_1 + x_2 + 2x_3$$

Block 1	Block 2	Block 3
000	100	200
112	212	012
210	010	110
120	220	020
022	122	222
202	002	102
221	021	121
101	201	001
011	111	211

The new design is a 180° rotation around the Factor B axis.

(b) Confound a 3^3 design in three blocks using the AB^2C component of the three-factor interaction. Compare your results with the design in Figure 9.7.

$$L = x_1 + 2x_2 + x_3$$

Block 1	Block 2	Block 3
000	210	112
022	202	120
011	221	101
212	100	010
220	122	002
201	111	021
110	012	200

102	020	222
121	001	211

The new design is a 180° rotation around the Factor *C* axis.

(c) Confound a 3^3 design three blocks using the *ABC* component of the three-factor interaction. Compare your results with the design in Figure 9.7.

$$L = x_1 + x_2 + x_3$$

Block 1	Block 2	Block 3
000	112	221
210	022	101
120	202	011
021	100	212
201	010	122
111	220	002
012	121	200
222	001	110
102	211	020

The new design is a 90° rotation around the Factor *C* axis along with switching layer 0 and layer 1 in the *C* axis.

(d) After looking at the designs in parts (a), (b), and (c) and Figure 9.7, what conclusions can you draw?

All four designs are relatively the same. The only differences are rotations and swapping of layers.

9.6. Confound a 3^4 design in three blocks using the AB^2CD component of the four-factor interaction.

The three blocks are shown below with $L = x_1 + 2x_2 + x_3 + x_4$

Block 1								
0000	1100	0110	0101	2200	0220	0202	1210	1201
0211	1222	2212	2221	0122	2111	1121	1112	2010
2102	0021	2001	2120	1011	2022	0012	1002	1020

Block 2								
1021	1110	1202	0001	0120	0212	1012	1101	1220
0200	0022	0111	2002	2121	2210	0010	0102	0221
1000	1122	1211	2112	2201	2020	2011	2100	2222

Block 3								
2012	2101	2220	1022	1111	1200	2000	2121	2211
1221	1010	1102	0020	0112	0201	1001	1120	1212
2021	2110	2202	0100	0222	0011	0002	0121	0210

9.7S. Consider the data from the first replicate of Problem 9.3. Assuming that all 27 observations could not be run on the same day, set up a design for conducting the experiment over three days with AB^2C confounded with blocks. Analyze the data.

Block 1			Block 2			Block 3		
000	=	3.45	100	=	4.07	200	=	4.20
110	=	4.38	210	=	4.26	010	=	4.14
011	=	5.22	111	=	5.15	211	=	5.17

102	=	4.30	202	=	4.17	002	=	4.08
201	=	4.96	001	=	4.80	101	=	4.52
212	=	4.86	012	=	3.94	112	=	4.53
121	=	6.25	221	=	6.03	021	=	6.21
022	=	5.14	122	=	4.99	222	=	4.85
220	=	5.67	020	=	5.80	120	=	5.48
Totals	=	44.23			43.21			43.18

The analysis of variance below identifies factors *B* and *C* as significant.

Design Expert Output

Response: Time

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	0.079	2	0.040			
Model	13.57	18	0.75	10.49	0.0040	significant
A	*0.11*	*2*	*0.055*	*0.77*	*0.5052*	
B	*8.42*	*2*	*4.21*	*58.63*	*0.0001*	
C	*3.81*	*2*	*1.91*	*26.53*	*0.0010*	
AB	*0.30*	*4*	*0.075*	*1.04*	*0.4573*	
AC	*0.11*	*4*	*0.028*	*0.39*	*0.8105*	
BC	*0.81*	*4*	*0.20*	*2.83*	*0.1234*	
Residual	0.43	6	0.072			
Cor Total	14.08	26				

The Model F-value of 10.49 implies the model is significant. There is only a 0.40% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case B, C are significant model terms.

9.8. Outline the analysis of variance table for the 3^4 design in nine blocks. Is this a practical design?

Source	DF
A	2
B	2
C	2
D	2
AB	4
AC	4
AD	4
BC	4
BD	4
CD	4
ABC (AB^2C,ABC^2,AB^2C^2)	6
ABD (ABD,AB^2D,ABD^2)	6
ACD (ACD,ACD^2,AC^2D^2)	6
BCD (BCD,BC^2D,BCD^2)	6
ABCD	16
Blocks (ABC,AB^2C^2,AC^2D,BC^2D^2)	8
Total	80

Any experiment with 81 runs is large. Instead of having three full levels of each factor, if two levels of each factor could be used, then the overall design would have 16 runs plus some center points. This two-level design could now probably be run in 2 or 4 blocks, with center points in each block. Additional curvature effects could be determined by augmenting the experiment with the axial points of a central composite design and additional enter points. The overall design would be less than 81 runs.

9.19S. Consider the data in Problem 9.3. If *ABC* is confounded in replicate I and ABC^2 is confounded in replicate II, perform the analysis of variance.

$L_1 = x_1 + x_2 + x_3$ $\qquad\qquad$ $L_2 = x_1 + x_2 + 2x_2$

Block 1			Block 2			Block 3			Block 1			Block 2			Block 3		
000	=	3.45	001	=	4.80	002	=	4.08	000	=	3.36	100	=	3.52	200	=	3.68
111	=	5.15	112	=	4.53	110	=	4.38	101	=	4.44	201	=	4.39	001	=	4.40
222	=	4.85	220	=	5.67	221	=	6.03	011	=	4.70	111	=	4.65	211	=	4.75
120	=	5.48	121	=	6.25	122	=	4.99	221	=	6.38	021	=	5.88	121	=	6.20
102	=	4.30	100	=	4.07	101	=	4.52	202	=	3.88	002	=	3.65	102	=	4.04
210	=	4.26	211	=	5.17	212	=	4.86	022	=	4.49	122	=	4.59	222	=	4.90
201	=	4.96	202	=	4.17	200	=	4.20	120	=	4.85	220	=	5.58	020	=	5.23
012	=	3.94	010	=	4.14	011	=	5.22	210	=	4.37	010	=	4.19	110	=	4.26
021	=	6.21	022	=	5.14	020	=	5.80	112	=	4.08	212	=	4.48	012	=	4.08

The sums of squares for *A, B, C, AB, AC,* and *BC* are calculated as usual. The only sums of squares presenting difficulties with calculations are the four components of the *ABC* interaction (*ABC*, ABC^2, AB^2C, and AB^2C^2). *ABC* is computed using replicate I and ABC^2 is computed using replicate II. AB^2C and AB^2C^2 are computed using data from both replicates.

We will show how to calculate AB^2C and AB^2C^2 from both replicates. Form a two-way table of *A* x *B* at each level of *C*. Find the *I*(*AB*) and *J*(*AB*) totals for each third of the *A* x *B* table.

		A				
C	*B*	0	1	2	I	J
	0	6.81	7.59	7.88	26.70	27.55
0	1	8.33	8.64	8.63	27.25	27.17
	2	11.03	10.33	11.25	26.54	25.77
	0	9.20	8.96	9.35	31.41	31.24
1	1	9.92	9.80	9.92	30.97	31.29
	2	12.09	12.45	12.41	31.72	31.57
	0	7.73	8.34	8.05	26.09	26.29
2	1	8.02	8.61	9.34	27.31	26.11
	2	9.63	9.58	9.75	25.65	26.65

The *I* and J components for each third of the above table are used to form a new table of diagonal totals.

C	*I*(*AB*)			*J*(*AB*)		
0	26.70	27.25	26.54	27.55	27.17	25.77
1	31.41	30.97	31.72	31.24	31.29	31.57
2	26.09	27.31	25.65	26.29	26.11	26.65

I Totals: 85.06, 85.26, 83.32 $\qquad$ *I* Totals: 85.49, 85.03, 83.12

J Totals: 85.73, 83.60, 84.31 $\qquad$ *J* Totals: 83.35,8 5.06, 85.23

$$\text{Now, } AB^2C^2 = I\left[CxI\left(AB\right)\right] = \frac{(85.06)^2 + (85.26)^2 + (83.32)^2}{18} - \frac{(253.64)^2}{54} = 0.1265$$

$$\text{and, } AB^2C = J\left[CxI\left(AB\right)\right] = \frac{(85.73)^2 + (83.60)^2 + (84.31)^2}{18} - \frac{(253.64)^2}{54} = 0.1307$$

If it were necessary, we could find ABC^2 as $ABC^2 = I[C \text{ x } J(AB)]$ and *ABC* as $J[C \text{ x } J(AB)]$. However, these components must be computed using the data from the appropriate replicate.

The analysis of variance table:

Source	*SS*	*DF*	*MS*	F_0
Replicates	1.06696	1		
Blocks within Replicates	0.2038	4		
A	0.4104	2	0.2052	5.02
B	17.7514	2	8.8757	217.0
C	7.6631	2	3.8316	93.68
AB	0.1161	4	0.0290	<1
AC	0.1093	4	0.0273	<1
BC	1.6790	4	0.4198	10.26
ABC (rep I)	0.0452	2	0.0226	<1
ABC^2 (rep II)	0.1020	2	0.0510	1.25
AB^2C	0.1307	2	0.0754	1.60
AB^2C^2	0.1265	2	0.0633	1.55
Error	0.8998	22	0.0409	
Total	30.3069	53		

9.10. Consider the data from replicate I in Problem 9.3. Suppose that only a one-third fraction of this design with $I=ABC$ is run. Construct the design, determine the alias structure, and analyze the design.

The design is 000, 012, 021, 102, 201, 111, 120, 210, 222.

The alias structure is:

$A = BC = AB^2C^2$
$B = AC = AB^2C$
$C = AB = ABC^2$
$AB^2 = AC^2 = BC^2$

		C		
A	*B*	0	1	2
0	0	3.45		
	1			5.48
	2		4.26	
1	0			6.21
	1		5.15	
	2	4.96		
2	0		3.94	
	1	4.30		
	2			4.85

Source	SS	DF
A	2.25	2
B	0.30	2
C	2.81	2
AB^2	0.30	2
Total	5.66	8

9.11. From examining Figure 9.9, what type of design would remain if after completing the first 9 runs, one of the three factors could be dropped?

The remaining design is a full 3^2 factorial.

9.12S. Construct a 3^{4-1}_{IV} design with $I=ABCD$. Write out the alias structure for this design.

The 27 runs for this design are as follows:

0000	1002	2001
0012	1011	2010
0021	1020	2022
0102	1101	2100
0111	1110	2112
0120	1122	2121
0201	1200	2202
0210	1212	2211
0222	1221	2220

$A = AB^2C^2D^2 = BCD$ $B = AB^2CD = ACD$ $C = ABC^2D = ABD$ $D = ABCD^2 = ABC$
$AB = ABC^2D^2 = CD$ $AB^2 = AC^2D^2 = BC^2D^2$ $AC = AB^2CD^2 = BD$ $AC^2 = AB^2D^2 = BC^2D$
$BC = AB^2C^2D = AD$ $BC^2 = AB^2D = AC^2D$ $BD^2 = AB^2C = ACD^2$ $CD^2 = ABC^2 = ABD^2$
$AD^2 = AB^2C^2 = BCD^2$

9.13. Verify that the design in Problem 9.12 is a resolution IV design.

The design in Problem 9.12 is a Resolution IV design because no main effect is aliased with a component of a two-factor interaction, but some two-factor interaction components are aliased with each other.

9.14S. Construct a 3^{5-2} design with $I=ABC$ and $I=CDE$. Write out the alias structure for this design. What is the resolution of this design?

The complete defining relation for this design is: $I = ABC = CDE = ABC^2DE = ABD^2E^2$
This is a resolution III design with $x_4 = 2x_1 + 2x_2 + x_3$ and $x_5 = x_1 + x_2 + 2x_4$ (mod 3).

00000	00112	00221
10022	10101	10210
20011	20120	20202
01022	01101	01210
11011	11120	11202
21000	21112	21221
02011	02120	02202
12000	12112	12221
22022	22101	22210

To find the alias of any effect, multiply the effect by I and I^2. For example, the alias of A is:

$A = AB^2C^2 = ACDE = AB^2CDE = AB^2DE = BC = AC^2D^2E^2 = BC^2DE = BD^2E^2$

9.15. Construct a 3^{9-6} design, and verify that it is a resolution III design.

Use the generators $I = AC^2D^2$, $I = AB^2C^2E$, $I = BC^2F^2$, $I = AB^2CG$, $I = ABCH^2$, and $I = ABJ^2$

000000000	021201102	102211001
022110012	212012020	001212210
011220021	100120211	211100110
221111221	122200220	020022222
210221200	010011111	222020101
202001212	201122002	200210122
112222112	002121120	121021010
101002121	111010202	110101022
120112100	220202011	012102201

To find the alias of any effect, multiply the effect by I and I^2. For example, the alias of C is:

$C = C(BC^2F^2) = BF^2$, At least one main effect is aliased with a component of a two-factor interaction.

9.16. Construct a 4 x 2^3 design confounded in two blocks of 16 observations each. Outline the analysis of variance for this design.

Design is a 4 x 2^3, with *ABC* at two levels, and *Z* at 4 levels. Represent *Z* with two pseudo-factors *D* and *E* as follows:

Factor *Z*	Pseudo-Factors *D*	*E*
Z_1	0	0 = (1)
Z_2	1	0 = *d*
Z_3	0	1 = *e*
Z_4	1	1 = *de*

The 4 x 2^3 is now a 2^5 in the factors *A, B, C, D* and *E*. Confound *ABCDE* with blocks. We have given both the letter notation and the digital notation for the treatment combinations.

Block 1			Block 2		
(1)	=	0000	*a*	=	1000
ab	=	1100	*b*	=	0100
ac	=	1010	*c*	=	0010
bc	=	0110	*abc*	=	1110
abcd	=	1111	*bcd*	=	0111
abce	=	1112	*bce*	=	0112
cd	=	0011	*acd*	=	1011
ce	=	0012	*ace*	=	1012
de	=	0003	*ade*	=	1003
abde	=	1103	*bde*	=	0103
bcde	=	0113	*abcde*	=	1113
be	=	0102	*abd*	=	1101
ad	=	1001	*abe*	=	1102
ae	=	1002	*d*	=	0001
acde	=	1013	*e*	=	0002
bd	=	0101	*cde*	=	0013

Source	*DF*
A	1
B	1
C	1
Z (D+E+DE)	3
AB	1
AC	1
AZ (AD+AE+ADE)	3
BC	1
BZ (BD+BE+BDE)	3
CZ (CD+CE+CDE)	3
ABC	1
ABZ (ABD+ABE+ABDE)	3
ACZ (ACD+ACE+ACDE)	3
BCZ (BCD+BCE+BCDE)	3
ABCZ (ABCD+ABCE)	2
Blocks (or *ABCDE*)	1
Total	31

9.17. Outline the analysis of variance table for a $2^2 3^2$ factorial design. Discuss how this design may be confounded in blocks.

Suppose we have n replicates of a $2^2 3^2$ factorial design. A and B are at 2 levels, and C and D are at 3 levels.

Source	*DF*	Components for Confounding
A	1	*A*
B	1	*B*
C	2	*C*
D	2	*D*
AB	1	*AB*
AC	2	*AC*
AD	2	*AD*
BC	2	*BD*
BD	2	CD,CD^2
CD	4	*ABC*
ABC	2	*ABD*
ABD	2	ACD,ACD^2
ACD	4	BCD,BCD^2
BCD	4	$ABCD,ABCD^2$
ABCD	4	
Error	36(n-1)	
Total	36n-1	

Confounding in this series of designs is discussed extensively by Margolin (1967). The possibilities for a single replicate of the $2^2 3^2$ design are:

2 blocks of 18 observations	4 blocks of 9 observations	9 blocks of 4 observations
3 blocks of 12 observations	6 blocks of 6 observations	

For example, one component of the four-factor interaction, say $ABCD^2$, could be selected to confound the design in 3 blocks of 12 observations each, while to confound the design in 2 blocks of 18 observations each we would select the *AB* interaction. Cochran and Cox (1957) and Anderson and McLean (1974) discuss confounding in these designs.

9.18. Starting with a 16-run 2^4 design, show how two three-level factors can be incorporated in this experiment. How many two-level factors can be included if we want some information on two-factor interactions?

Use column *A* and *B* for one three-level factor and columns *C* and *D* for the other. Use the *AC* and *BD* columns for the two, two-level factors. The design will be of resolution V.

9.19. Starting with a 16-run 2^4 design, show how one three-level factor and three two-level factors can be accommodated and still allow the estimation of two-factor interactions.

Use columns *A* and *B* for the three-level factor, and columns *C* and *D* and *ABCD* for the three two-level factors. This design will be of resolution V.

9.20. In Problem 8.24, you met Harry Peterson-Nedry, a friend of the author who has a winery and vineyard in Newberg, Oregon. That problem described the application of two-level fractional factorial designs to their 1985 Pinor Noir product. In 1987, they wanted to conduct another Pinot Noir experiment. The variables for this experiment were

Variable	Levels
Clone of Pinot Noir	Wadenswil, Pommard
Berry Size	Small, Large
Fermentation temperature	80F, 85F, 90/80F, 90F
Whole Berry	None, 10%
Maceration Time	10 days, 21 days
Yeast Type	Assmanhau, Champagne
Oak Type	Troncais, Allier

Harry decided to use a 16-run two-level fractional factorial design, treating the four levels of fermentation temperature as two two-level variables. As in the Chapter 8 Problem, the rankings from a taste-test panel was the response variable. The design and the resulting average ranks are shown below:

Run	Clone	Size	Berry Temp.	Ferm. Berry	Whole Time	Macer. Type	Yeast Type	Oak Rank	Average
1	–	–	–	–	–	–	–	–	4
2	+	–	–	–	–	+	+	+	10
3	–	+	–	–	+	–	+	+	6
4	+	+	–	–	+	+	–	–	9
5	–	–	+	–	+	+	+	–	11
6	+	–	+	–	+	–	–	+	1
7	–	+	+	–	–	+	–	+	15
8	+	+	+	–	–	–	+	–	5
9	–	–	–	+	+	+	–	+	12
10	+	–	–	+	+	–	+	–	2
11	–	+	–	+	–	+	+	–	16
12	+	+	–	+	–	–	–	+	3
13	–	–	+	+	–	–	+	+	8
14	+	–	+	+	–	+	–	–	14
15	–	+	+	+	+	–	–	–	7
16	+	+	+	+	+	+	+	+	13

(a) Describe the aliasing in this design.

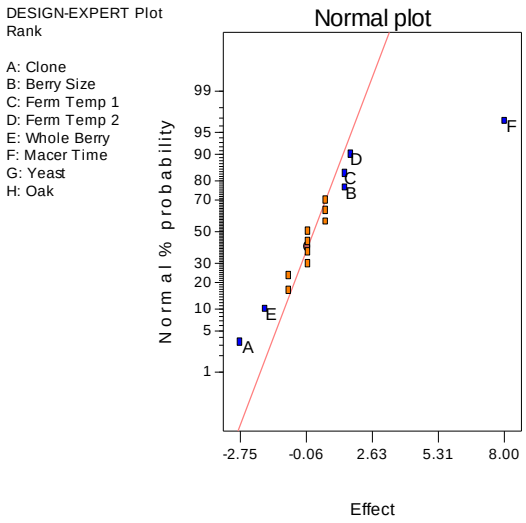
The design is a resolution IV design such that the main effects are aliased with three factor interactions.

Design Expert Output

Term	Aliases
Intercept	ABCG ABDH ABEF ACDF ACEH ADEG AFGH BCDE BCFH BDFG BEGH CDGH CEFG DEFH
A	BCG BDH BEF CDF CEH DEG FGH ABCDE
B	ACG ADH AEF CDE CFH DFG EGH
C	ABG ADF AEH BDE BFH DGH EFG
D	ABH ACF AEG BCE BFG CGH EFH
E	ABF ACH ADG BCD BGH CFG DFH
F	ABE ACD AGH BCH BDG CEG DEH
G	ABC ADE AFH BDF BEH CDH CEF
H	ABD ACE AFG BCF BEG CDG DEF
AB	CG DH EF ACDE ACFH ADFG AEGH BCDF BCEH BDEG BFGH
AC	BG DF EH ABDE ABFH ADGH AEFG BCDH BCEF CDEG CFGH
AD	BH CF EG ABCE ABFG ACGH AEFH BCDG BDEF CDEH DFGH
AE	BF CH DG ABCD ABGH ACFG ADFH BCEG BDEH CDEF EFGH
AF	BE CD GH ABCH ABDG ACEG ADEH BCFG BDFH CEFH DEFG
AG	BC DE FH ABDF ABEH ACDH ACEF BDGH BEFG CDFG CEGH
AH	BD CE FG ABCF ABEG ACDG ADEF BCGH BEFH CDFH DEGH

(b) Analyze the data and draw conclusions.

All of the main effects except Yeast and Oak are significant. The Maceration Time is the most significant. None of the interactions were significant.



Design Expert Output

Response: Rank

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	328.75	6	54.79	43.83	< 0.0001	significant
A	*30.25*	*1*	*30.25*	*24.20*	*0.0008*	
B	*9.00*	*1*	*9.00*	*7.20*	*0.0251*	
C	*9.00*	*1*	*9.00*	*7.20*	*0.0251*	
D	*12.25*	*1*	*12.25*	*9.80*	*0.0121*	
E	*12.25*	*1*	*12.25*	*9.80*	*0.0121*	
F	*256.00*	*1*	*256.00*	*204.80*	*< 0.0001*	
Residual	11.25	9	1.25			
Cor Total	340.00	15				

The Model F-value of 43.83 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.12	R-Squared	0.9669
Mean	8.50	Adj R-Squared	0.9449
C.V.	13.15	Pred R-Squared	0.8954

PRESS	35.56	Adeq Precision	19.270		

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	8.50	1	0.28	7.87	9.13	
A-Clone	-1.38	1	0.28	-2.01	-0.74	1.00
B-Berry Size	0.75	1	0.28	0.12	1.38	1.00
C-Ferm Temp 1	0.75	1	0.28	0.12	1.38	1.00
D-Ferm Temp 2	0.88	1	0.28	0.24	1.51	1.00
E-Whole Berry	-0.87	1	0.28	-1.51	-0.24	1.00
F-Macer Time	4.00	1	0.28	3.37	4.63	1.00

Final Equation in Terms of Coded Factors:

Rank =
+8.50
-1.38 * A
+0.75 * B
+0.75 * C
+0.88 * D
-0.87 * E
+4.00 * F

(c) What comparisons can you make between this experiment and the 1985 Pinot Noir experiment from Problem 8.30?

The experiment from Chapter 8 indicates that Yeast, Barrel, Whole Cluster and the Clone X Yeast interactions were significant. This experiment indicates that Maceration Time, Whole Berry, Clone and Fermentation Temperature are significant.

9.21. Suppose there are four three-level categorical factors and a style two-level continuous factor. What is the minimum number of runs required to estimate all main effects and two-factor interactions? Construct this design.

A minimum of 42 runs are required to estimate all main effects and two factor interactions. Below is a *D*-optimal design with 42 runs.

Run	Factor *A*	Factor *B*	Factor *C*	Factor *D*	Factor *E*
1	2	1	1	1	-1
2	1	1	2	1	+1
3	3	3	2	3	-1
4	2	2	1	3	-1
5	1	1	1	2	+1
6	3	2	3	3	-1
7	1	2	1	2	-1
8	1	1	3	2	+1
9	2	1	3	2	+1
10	1	2	2	2	+1
11	2	3	3	1	-1
12	3	2	2	3	+1
13	2	1	3	1	-1
14	3	1	1	2	+1
15	2	3	1	2	-1
16	3	1	3	3	+1
17	2	2	2	1	-1
18	3	3	1	2	+1
19	1	1	3	3	-1
20	3	3	3	2	-1
21	3	2	1	1	-1
22	1	3	3	3	-1

23	1	3	2	3	+1
24	2	1	2	2	-1
25	1	3	2	2	-1
26	3	2	3	1	+1
27	1	3	1	1	-1
28	2	2	3	3	+1
29	1	3	3	1	+1
30	3	1	1	3	-1
31	2	3	2	2	+1
32	1	2	3	1	-1
33	3	2	2	2	-1
34	2	3	1	1	+1
35	3	1	2	1	-1
36	3	3	2	1	+1
37	1	2	1	3	+1
38	2	1	2	3	-1
39	3	1	1	1	+1
40	2	3	3	3	+1
41	1	2	2	3	-1
42	2	2	1	2	+1

9.22S. Reconsider the experiment in Problem 9.21. Construct a design with $N = 48$ runs and compare it to the design you constructed in Problem 9.21.

When generating the *D*-optimal design in Design Expert software, an additional six runs were included for lack of fit. The resulting design is shown below. As with the design created in Problem 9.21, the three level factors are balanced with equal runs at each individual factor level, and the two level factor is very slightly unbalanced with 25 low level runs and 23 high level runs.

Run	Factor *A*	Factor *B*	Factor *C*	Factor *D*	Factor *E*
1	1	1	3	3	-1
2	3	3	1	1	-1
3	2	3	3	3	+1
4	3	1	2	1	-1
5	1	2	3	2	+1
6	2	1	2	3	-1
7	2	2	3	3	-1
8	3	1	2	2	+1
9	2	2	1	1	+1
10	1	3	3	3	+1
11	2	2	1	2	-1
12	3	2	2	3	-1
13	1	3	1	2	-1
14	2	3	1	2	+1
15	3	1	3	1	-1
16	1	2	2	3	+1
17	3	3	3	3	-1
18	1	2	2	1	+1
19	1	1	1	2	+1
20	3	3	2	3	+1
21	2	2	3	1	-1
22	3	3	3	2	+1
23	2	1	1	1	-1
24	1	2	1	1	-1
25	3	1	3	3	+1
26	1	3	2	3	-1
27	3	1	1	1	+1

28	1	3	3	1	-1
29	2	1	3	2	-1
30	1	3	2	2	+1
31	3	2	3	2	-1
32	2	3	2	1	+1
33	1	1	2	2	-1
34	2	1	1	3	+1
35	2	2	2	2	+1
36	2	3	2	2	-1
37	3	1	1	3	-1
38	1	1	2	1	+1
39	3	2	1	2	+1
40	1	3	1	1	+1
41	2	3	1	3	-1
42	3	2	3	1	+1
43	2	1	3	1	+1
44	1	2	1	2	-1
45	3	2	2	3	+1
46	3	1	1	1	-1
47	1	2	2	2	-1
48	2	3	3	3	-1

9.23. Reconsider the experiment in Problem 9.21. Suppose that you are only interested in main effects. Construct a design with $N = 12$ runs for this experiment.

Run	Factor *A*	Factor *B*	Factor *C*	Factor *D*	Factor *E*
1	1	3	2	3	+1
2	1	2	3	2	-1
3	2	2	1	3	+1
4	1	1	1	1	-1
5	3	2	2	1	-1
6	3	1	3	3	-1
7	3	1	1	2	+1
8	3	3	3	1	+1
9	2	1	3	1	+1
10	2	3	2	2	-1
11	2	2	1	2	-1
12	1	3	2	1	-1

9.24. An article in the *Journal of Chemical Technology and Biotechnology* ("A Study of Antifungal Antibiotic Production by *Thermomomospora* sp MTCC 3340 Using Full Factorial Design," 2003, Vol. 78, pp. 605-610) investigated three independent variables – concentration of carbon source (glucose), concentration of nitrogen source (soybean meal), and temperature of incubation for their effects on the production of antifungal antibiotic by the isolate *Thermomonospora* sp MTCC 3340. A 3^3 factorial design was conducted and the results are shown in the table in Table P9.1.

Run	% Carbon	% Nitrogen	Temp	Activity
1	2	0.5	25	25.84
2	2	1	25	51.86
3	2	3	25	32.59
4	5	0.5	25	20.48
5	5	1	25	25.84
6	5	3	25	12.87
7	7.5	0.5	25	20.48
8	7.5	1	25	25.84

9	7.5	3	25	10.2
10	2	0.5	30	51.86
11	2	1	30	131.33
12	2	3	30	41.11
13	5	0.5	30	41.11
14	5	1	30	104.11
15	5	3	30	32.59
16	7.5	0.5	30	65.42
17	7.5	1	30	82.53
18	7.5	3	30	51.86
19	2	0.5	35	41.11
20	2	1	35	104.11
21	2	3	35	32.59
22	5	0.5	35	32.59
23	5	1	35	82.53
24	5	3	35	25.84
25	7.5	0.5	35	51.86
26	7.5	1	35	65.42
27	7.5	3	35	41.11

(a) Analyze the data from this experiment.

The ANOVA is shown below. Factor *B*, and the B^2 and C^2 quadratic terms appear to be significant.

Design Expert Output

Response 1 Activity

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	19338.43	9	2148.71	8.63	< 0.0001	significant
A-Carbon	*433.77*	*1*	*433.77*	*1.74*	*0.2043*	
B-Nitrogen	*280.10*	*1*	*280.10*	*1.13*	*0.3036*	
C-Temperature	*2913.48*	*1*	*2913.48*	*11.70*	*0.0033*	
AB	*50.78*	*1*	*50.78*	*0.20*	*0.6572*	
AC	*97.05*	*1*	*97.05*	*0.39*	*0.5407*	
BC	*215.04*	*1*	*215.04*	*0.86*	*0.3657*	
A2	*486.58*	*1*	*486.58*	*1.95*	*0.1801*	
B2	*7503.07*	*1*	*7503.07*	*30.14*	*< 0.0001*	
C2	*4642.23*	*1*	*4642.23*	*18.65*	*0.0005*	
Residual	4231.91	17	248.94			
Cor Total	23570.33	26				

Std. Dev.	15.78	R-Squared	0.8205
Mean	48.34	Adj R-Squared	0.7254
C.V. %	32.64	Pred R-Squared	0.5795
PRESS	9911.08	Adeq Precision	10.086

(b) Fit a second-order model to the activity response. Construct contour plots and response surface plots to assist in interpreting the results of this experiment.

The model, contour plot, and response surface plot are shown below. Because factor *A* was not significant, the plots are shown with factors *B* and *C* only.

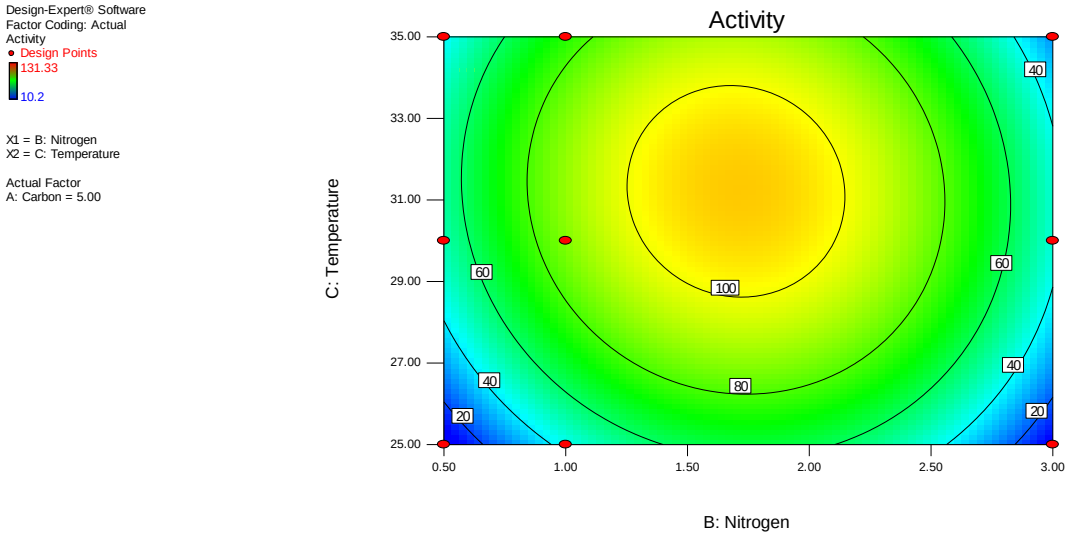
Design Expert Output

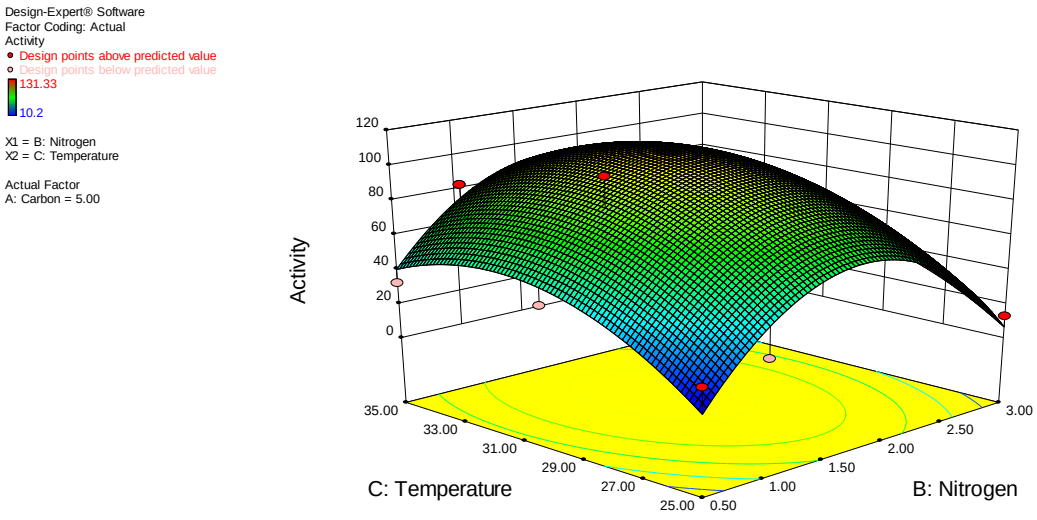
Term	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF

Intercept	106.17	1	11.02	82.91	129.43	
A-Carbon	-5.04	1	3.82	-13.09	3.01	1.06
B-Nitrogen	-3.95	1	3.72	-11.80	3.90	1.12
C-Temperature	13.07	1	3.82	5.01	21.13	1.05
AB	1.94	1	4.30	-7.13	11.01	1.05
AC	2.84	1	4.55	-6.76	12.44	1.00
BC	-4.00	1	4.30	-13.08	5.08	1.05
A2	9.09	1	6.50	-4.63	22.81	1.00
B2	-58.48	1	10.65	-80.95	-36.00	1.12
C2	-27.82	1	6.44	-41.41	-14.23	1.00

Final Equation in Terms of Actual Factors:

Activity =
-1046.32378
-20.43903 * Carbon
+144.34481 * Nitrogen
+69.50973 * Temperature
+0.56468 * Carbon * Nitrogen
+0.20654 * Carbon * Temperature
-0.64000 * Nitrogen * Temperature
+1.20237 * Carbon2
-37.42422 * Nitrogen2
-1.11262 * Temperature2





(c) What operating conditions would you recommend to optimize this process?

Set factor *B*, Nitrogen, at approximately 1.7, and factor *C*, Temperature, at 31, maximizes the Activity. Factor *A* is not significant, so any level can be chosen depending on the manufacturing preferences.

9.25. Construct a minimum run *D*-optimal resolution IV design for 10 factors. Find the alias relationships. What approach would you recommend for analyzing the data from this experiment?

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	*K*
–	+	+	–	–	–	+	+	–	–
+	+	+	+	+	–	+	+	+	+
–	+	–	–	+	+	–	+	+	–
–	+	–	+	–	–	–	+	–	+
+	+	–	–	–	–	–	–	+	–
+	+	–	–	+	+	+	–	–	+
–	–	–	–	+	–	+	–	+	+
+	–	–	+	+	–	+	+	–	–
–	+	+	+	+	–	–	–	–	–
–	+	–	+	–	+	+	–	+	–
+	–	–	+	+	+	–	–	+	+
–	–	–	–	–	+	–	–	–	–
+	–	+	+	–	–	+	–	–	+
+	–	+	–	+	+	+	–	+	–
–	–	+	+	+	+	+	+	–	+
–	–	+	+	–	–	–	+	+	–
+	+	+	+	–	+	–	+	–	–
–	+	+	–	–	+	–	–	+	+
+	–	–	–	–	+	+	+	+	+
+	–	+	–	+	–	–	+	–	+

The alias relationships for the main effects and two factor interactions are as follows:

[Intercept] = Intercept - 0.333 * *BE* - 0.333 * *BG* - 0.333 * *BK* + 0.333 * *CD* + 0.333 * *CH* - 0.333 * *CJ* + 0.333 * *DH* - 0.333 * *DJ* + 0.333 * *EG* + 0.333 * *EK* + 0.333 * *GK* - 0.333 * *HJ*

[*A*] = *A* [*B*] = *B* [*C*] = *C* [*D*] = *D* [*E*] = *E*
[*F*] = *F* [*G*] = *G* [*H*] = *H* [*J*] = *J* [*K*] = *K*

[*AB*] = *AB* + 0.333 * *BE* + 0.333 * *BG* + 0.333 * *BK* + 0.667 * *CD* + 0.667 * *CH* + 0.333 * *CJ* + 0.667 * *DH* + 0.333 * *DJ* + 0.667 * *EG* + 0.667 * *EK* - *FJ* + 0.667 * *GK* + 0.333 * *HJ*

[*AC*] = *AC* - 0.333 * *BC* + 0.667 * *BD* + 0.667 * *BH* + 0.333 * *CE* + 0.333 * *CG* - 0.667 * *DE* + 0.333 * *DK* + 0.667 * *EJ* - *FK* - 0.667 * *GH* + 0.667 * *GJ* + 0.333 * *HK* - 0.333 * *JK*

[*AD*] = *AD* + 0.667 * *BC* - 0.333 * *BD* + 0.667 * *BH* - 0.667 * *CE* + 0.333 * *CG* + 0.333 * *DE* + 0.333 * *DK* + 0.667 * *EJ* - *FG* + 0.333 * *GH* - 0.333 * *GJ* - 0.667 * *HK* + 0.667 * *JK*

[*AE*] = *AE* - 0.333 * *BE* + 0.667 * *BG* + 0.667 * *BK* - 0.667 * *CD* + 0.333 * *CH* + 0.667 * *CJ* + 0.333 * *DH* + 0.667 * *DJ* + 0.333 * *EG* + 0.333 * *EK* - *FH* - 0.667 * *GK* - 0.333 * *HJ*

[*AF*] = *AF* + 0.333 * *BC* + 0.333 * *BD* + 0.333 * *BH* - *BJ* - 0.333 * *CE* - 0.333 * *CG* - *CK* - 0.333 * *DE* - *DG* - 0.333 * *DK* - *EH* + 0.333 * *EJ* - 0.333 * *GH* + 0.333 * *GJ* - 0.333 * *HK* + 0.333 * *JK*

[*AG*] = *AG* + 0.667 * *BE* - 0.333 * *BG* + 0.667 * *BK* + 0.333 * *CD* - 0.667 * *CH* + 0.667 * *CJ* - *DF* + 0.333 * *DH* - 0.333 * *DJ* + 0.333 * *EG* - 0.667 * *EK* + 0.333 * *GK* + 0.667 * *HJ*

[*AH*] = *AH* + 0.667 * *BC* + 0.667 * *BD* - 0.333 * *BH* + 0.333 * *CE* - 0.667 * *CG* + 0.333 * *DE* - 0.667 * *DK* - *EF* - 0.333 * *EJ* + 0.333 * *GH* + 0.667 * *GJ* + 0.333 * *HK* + 0.667 * *JK*

[*AJ*] = *AJ* + 0.333 * *BC* + 0.333 * *BD* - *BF* + 0.333 * *BH* + 0.667 * *CE* + 0.667 * *CG* + 0.667 * *DE* + 0.667 * *DK* + 0.333 * *EJ* + 0.667 * *GH* + 0.333 * *GJ* + 0.667 * *HK* + 0.333 * *JK*

[*AK*] = *AK* + 0.667 * *BE* + 0.667 * *BG* - 0.333 * *BK* + 0.333 * *CD* - *CF* + 0.333 * *CH* - 0.333 * *CJ* - 0.667 * *DH* + 0.667 * *DJ* - 0.667 * *EG* + 0.333 * *EK* + 0.333 * *GK* + 0.667 * *HJ*

Forward selection regression should be used for analyzing the experiment. Hierarchy should be considered.

9.26. Construct a minimum run *D*-optimal resolution IV design for 12 factors. Find the alias relationships. What approach would you recommend for analyzing the data from this experiment?

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*	*K*	*L*	*M*
–	–	+	–	–	+	+	–	–	–	–	+
–	–	–	–	+	+	–	–	+	+	+	+
–	+	–	+	+	+	+	+	–	+	–	+
–	+	–	–	–	–	–	+	+	–	–	+
+	+	–	+	–	+	+	–	+	–	+	+
–	+	+	–	–	+	+	+	+	+	+	–
–	–	–	+	–	+	–	+	–	–	+	–
–	–	+	+	+	–	+	+	+	–	+	+
+	–	–	–	+	+	+	+	+	–	–	–
–	–	–	+	–	–	+	–	+	+	–	–

+	–	–	–	–	–	+	+	–	+	+	+
–	+	–	–	+	–	+	–	–	–	+	–
+	+	–	+	+	–	–	+	+	+	+	–
+	+	+	+	–	–	+	+	–	–	–	–
+	–	+	–	–	–	–	–	+	–	+	–
+	–	+	+	+	+	+	–	–	+	+	–
–	–	+	–	+	–	–	+	–	+	–	–
+	–	–	+	+	–	–	–	–	–	–	+
+	+	+	–	+	–	+	–	+	+	–	+
+	+	–	–	–	+	–	–	–	+	–	–
–	+	+	+	–	–	–	–	–	+	+	+
+	+	+	–	+	+	–	+	–	–	+	+
–	+	+	+	+	+	–	–	+	–	–	–
+	–	+	+	–	+	–	+	+	+	–	+

The alias relationships for the main effects and two factor interactions are as follows:

[Intercept] = Intercept

[*A*] = *A* [*B*] = *B* [*C*] = *C* [*D*] = *D* [*E*] = *E*
[*F*] = *F* [*G*] = *G* [*H*] = *H* [*J*] = *J* [*K*] = *K*
[*L*] = *L* [*M*] = *M*

[*AB*] = *AB* - 0.333 * *CD* + 0.333 * *CE* - 0.333 * *CF* + 0.333 * *CG* + 0.333 * *CH*
- 0.333 * *CJ* - 0.333 * *CK* - 0.333 * *CL* + 0.333 * *CM* - 0.333 * *DE* - 0.333 * *DF*
+ 0.333 * *DG* + 0.333 * *DH* + 0.333 * *DJ* - 0.333 * *DK* + 0.333 * *DL* - 0.333 * *DM*
- 0.333 * *EF* - 0.333 * *EG* + 0.333 * *EH* + 0.333 * *EJ* + 0.333 * *EK* + 0.333 * *EL*
+ 0.333 * *EM* - 0.333 * *FG* - 0.333 * *FH* - 0.333 * *FJ* - 0.333 * *FK* + 0.333 * *FL*
+ 0.333 * *FM* - 0.333 * *GH* + 0.333 * *GJ* - 0.333 * *GK* - 0.333 * *GL* + 0.333 * *GM*
- 0.333 * *HJ* - 0.333 * *HK* + 0.333 * *HL* - 0.333 * *HM* + 0.333 * *JK* + 0.333 * *JL*
+ 0.333 * JM - 0.333 * KL - 0.333 * KM + 0.333 * LM

[*AC*] = *AC* - 0.333 * *BD* + 0.333 * *BE* - 0.333 * *BF* + 0.333 * *BG* + 0.333 * *BH*
- 0.333 * *BJ* - 0.333 * *BK* - 0.333 * *BL* + 0.333 * *BM* - 0.333 * *DE* + 0.333 * *DF*
+ 0.333 * *DG* + 0.333 * *DH* - 0.333 * *DJ* + 0.333 * *DK* - 0.333 * *DL* - 0.333 * *DM*
+ 0.333 * *EF* + 0.333 * *EG* - 0.333 * *EH* - 0.333 * *EJ* + 0.333 * *EK* + 0.333 * *EL*
+ 0.333 * *EM* - 0.333 * *FG* + 0.333 * *FH* - 0.333 * *FJ* + 0.333 * *FK* + 0.333 * *FL*
+ 0.333 * *FM* - 0.333 * *GH* - 0.333 * *GJ* + 0.333 * *GK* - 0.333 * *GL* - 0.333 * *GM*
- 0.333 * *HJ* - 0.333 * *HK* - 0.333 * *HL* + 0.333 * *HM* + 0.333 * *JK* - 0.333 * *JL*
+ 0.333 * *JM* - 0.333 * *KL* + 0.333 * *KM* - 0.333 * *LM*

[*AD*] = *AD* - 0.333 * *BC* - 0.333 * *BE* - 0.333 * *BF* + 0.333 * *BG* + 0.333 * *BH*
+ 0.333 * *BJ* - 0.333 * *BK* + 0.333 * *BL* - 0.333 * *BM* - 0.333 * *CE* + 0.333 * *CF*
+ 0.333 * *CG* + 0.333 * *CH* - 0.333 * *CJ* + 0.333 * *CK* - 0.333 * *CL* - 0.333 * *CM*
- 0.333 * *EF* - 0.333 * *EG* - 0.333 * *EH* - 0.333 * *EJ* + 0.333 * *EK* + 0.333 * *EL*
- 0.333 * *EM* + 0.333 * *FG* - 0.333 * *FH* + 0.333 * *FJ* + 0.333 * *FK* + 0.333 * *FL*
+ 0.333 * *FM* - 0.333 * *GH* - 0.333 * *GJ* - 0.333 * *GK* + 0.333 * *GL* - 0.333 * *GM*
+ 0.333 * *HJ* + 0.333 * *HK* - 0.333 * *HL* - 0.333 * *HM* + 0.333 * *JK* + 0.333 * *JL*
+ 0.333 * *JM* + 0.333 * *KL* - 0.333 * *KM* - 0.333 * *LM*

[*AE*] = *AE* + 0.333 * *BC* - 0.333 * *BD* - 0.333 * *BF* - 0.333 * *BG* + 0.333 * *BH*
+ 0.333 * *BJ* + 0.333 * *BK* + 0.333 * *BL* + 0.333 * *BM* - 0.333 * *CD* + 0.333 * *CF*
+ 0.333 * *CG* - 0.333 * *CH* - 0.333 * *CJ* + 0.333 * *CK* + 0.333 * *CL* + 0.333 * *CM*
- 0.333 * *DF* - 0.333 * *DG* - 0.333 * *DH* - 0.333 * *DJ* + 0.333 * *DK* + 0.333 * *DL*
- 0.333 * *DM* + 0.333 * *FG* + 0.333 * *FH* - 0.333 * *FJ* - 0.333 * *FK* + 0.333 * *FL*
- 0.333 * *FM* - 0.333 * *GH* + 0.333 * *GJ* + 0.333 * *GK* - 0.333 * *GL* - 0.333 * *GM*
+ 0.333 * *HJ* - 0.333 * *HK* + 0.333 * *HL* - 0.333 * *HM* + 0.333 * *JK* - 0.333 * *JL*

$$- 0.333 * JM + 0.333 * KL - 0.333 * KM - 0.333 * LM$$

$$[AF] = AF - 0.333 * BC - 0.333 * BD - 0.333 * BE - 0.333 * BG - 0.333 * BH - 0.333 * BJ - 0.333 * BK + 0.333 * BL + 0.333 * BM + 0.333 * CD + 0.333 * CE - 0.333 * CG + 0.333 * CH - 0.333 * CJ + 0.333 * CK + 0.333 * CL + 0.333 * CM - 0.333 * DE + 0.333 * DG - 0.333 * DH + 0.333 * DJ + 0.333 * DK + 0.333 * DL + 0.333 * DM + 0.333 * EG + 0.333 * EH - 0.333 * EJ - 0.333 * EK + 0.333 * EL - 0.333 * EM - 0.333 * GH + 0.333 * GJ - 0.333 * GK + 0.333 * GL - 0.333 * GM + 0.333 * HJ - 0.333 * HK - 0.333 * HL + 0.333 * HM - 0.333 * JK - 0.333 * JL + 0.333 * JM - 0.333 * KL - 0.333 * KM + 0.333 * LM$$

$$[AG] = AG + 0.333 * BC + 0.333 * BD - 0.333 * BE - 0.333 * BF - 0.333 * BH + 0.333 * BJ - 0.333 * BK - 0.333 * BL + 0.333 * BM + 0.333 * CD + 0.333 * CE - 0.333 * CF - 0.333 * CH - 0.333 * CJ + 0.333 * CK - 0.333 * CL - 0.333 * CM - 0.333 * DE + 0.333 * DF - 0.333 * DH - 0.333 * DJ - 0.333 * DK + 0.333 * DL - 0.333 * DM + 0.333 * EF - 0.333 * EH + 0.333 * EJ + 0.333 * EK - 0.333 * EL - 0.333 * EM - 0.333 * FH + 0.333 * FJ - 0.333 * FK + 0.333 * FL - 0.333 * FM - 0.333 * HJ - 0.333 * HK - 0.333 * HL - 0.333 * HM - 0.333 * JK - 0.333 * JL + 0.333 * JM + 0.333 * KL + 0.333 * KM + 0.333 * LM$$

$$[AH] = AH + 0.333 * BC + 0.333 * BD + 0.333 * BE - 0.333 * BF - 0.333 * BG - 0.333 * BJ - 0.333 * BK + 0.333 * BL - 0.333 * BM + 0.333 * CD - 0.333 * CE + 0.333 * CF - 0.333 * CG - 0.333 * CJ - 0.333 * CK - 0.333 * CL + 0.333 * CM - 0.333 * DE - 0.333 * DF - 0.333 * DG + 0.333 * DJ + 0.333 * DK - 0.333 * DL - 0.333 * DM + 0.333 * EF - 0.333 * EG + 0.333 * EJ - 0.333 * EK + 0.333 * EL - 0.333 * EM - 0.333 * FG + 0.333 * FJ - 0.333 * FK - 0.333 * FL + 0.333 * FM - 0.333 * GJ - 0.333 * GK - 0.333 * GL - 0.333 * GM + 0.333 * JK - 0.333 * JL - 0.333 * JM + 0.333 * KL + 0.333 * KM + 0.333 * LM$$

$$[AJ] = AJ - 0.333 * BC + 0.333 * BD + 0.333 * BE - 0.333 * BF + 0.333 * BG - 0.333 * BH + 0.333 * BK + 0.333 * BL + 0.333 * BM - 0.333 * CD - 0.333 * CE - 0.333 * CF - 0.333 * CG - 0.333 * CH + 0.333 * CK - 0.333 * CL + 0.333 * CM - 0.333 * DE + 0.333 * DF - 0.333 * DG + 0.333 * DH + 0.333 * DK + 0.333 * DL + 0.333 * DM - 0.333 * EF + 0.333 * EG + 0.333 * EH + 0.333 * EK - 0.333 * EL - 0.333 * EM + 0.333 * FG + 0.333 * FH - 0.333 * FK - 0.333 * FL + 0.333 * FM - 0.333 * GH - 0.333 * GK - 0.333 * GL + 0.333 * GM + 0.333 * HK - 0.333 * HL - 0.333 * HM - 0.333 * KL + 0.333 * KM - 0.333 * LM$$

$$[AK] = AK - 0.333 * BC - 0.333 * BD + 0.333 * BE - 0.333 * BF - 0.333 * BG - 0.333 * BH + 0.333 * BJ - 0.333 * BL - 0.333 * BM + 0.333 * CD + 0.333 * CE + 0.333 * CF + 0.333 * CG - 0.333 * CH + 0.333 * CJ - 0.333 * CL + 0.333 * CM + 0.333 * DE + 0.333 * DF - 0.333 * DG + 0.333 * DH + 0.333 * DJ + 0.333 * DL - 0.333 * DM - 0.333 * EF + 0.333 * EG - 0.333 * EH + 0.333 * EJ + 0.333 * EL - 0.333 * EM - 0.333 * FG - 0.333 * FH - 0.333 * FJ - 0.333 * FL - 0.333 * FM - 0.333 * GH - 0.333 * GJ + 0.333 * GL + 0.333 * GM + 0.333 * HJ + 0.333 * HL + 0.333 * HM - 0.333 * JL + 0.333 * JM - 0.333 * LM$$

$$[AL] = AL - 0.333 * BC + 0.333 * BD + 0.333 * BE + 0.333 * BF - 0.333 * BG + 0.333 * BH + 0.333 * BJ - 0.333 * BK + 0.333 * BM - 0.333 * CD + 0.333 * CE + 0.333 * CF - 0.333 * CG - 0.333 * CH - 0.333 * CJ - 0.333 * CK - 0.333 * CM + 0.333 * DE + 0.333 * DF + 0.333 * DG - 0.333 * DH + 0.333 * DJ + 0.333 * DK - 0.333 * DM + 0.333 * EF - 0.333 * EG + 0.333 * EH - 0.333 * EJ + 0.333 * EK - 0.333 * EM + 0.333 * FG - 0.333 * FH - 0.333 * FJ - 0.333 * FK + 0.333 * FM - 0.333 * GH - 0.333 * GJ + 0.333 * GK + 0.333 * GM - 0.333 * HJ + 0.333 * HK + 0.333 * HM - 0.333 * JK - 0.333 * JM - 0.333 * KM$$

$$[AM] = AM + 0.333 * BC - 0.333 * BD + 0.333 * BE + 0.333 * BF + 0.333 * BG - 0.333 * BH + 0.333 * BJ - 0.333 * BK + 0.333 * BL - 0.333 * CD + 0.333 * CE + 0.333 * CF - 0.333 * CG + 0.333 * CH + 0.333 * CJ + 0.333 * CK - 0.333 * CL - 0.333 * DE + 0.333 * DF - 0.333 * DG - 0.333 * DH + 0.333 * DJ - 0.333 * DK - 0.333 * DL - 0.333 * EF - 0.333 * EG - 0.333 * EH - 0.333 * EJ - 0.333 * EK - 0.333 * EL - 0.333 * FG + 0.333 * FH + 0.333 * FJ - 0.333 * FK + 0.333 * FL - 0.333 * GH + 0.333 * GJ + 0.333 * GK + 0.333 * GL - 0.333 * HJ + 0.333 * HK + 0.333 * HL + 0.333 * JK - 0.333 * JL - 0.333 * KL$$

Forward selection regression should be used for analyzing the experiment. Hierarchy should be considered.

9.27. Suppose that you must design an experiment to investigate nine continuous factors. It is thought that running all factors at two levels is adequate but that all two-factor interactions are of interest.

(a) How many runs are required to estimate all main effects and two factor-interactions?

1 for the intercept term
9 for the main effects
$36\left(\frac{9!}{2!(9-2)!}\right)$ for the two-factor interactions
Total of 46 runs.

(b) Find a minimum-run *D*-optimal design that is suitable for this problem.

A	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*J*
+	–	–	+	+	–	–	–	–
+	–	+	–	+	–	+	+	+
+	+	–	+	+	–	–	+	+
+	+	+	–	+	+	+	–	+
+	+	–	–	+	–	–	–	+
–	–	–	–	+	+	–	+	+
+	–	+	+	+	+	–	+	+
+	+	+	+	+	+	–	–	–
+	–	–	+	–	–	+	–	+
+	–	–	+	–	+	–	+	–
–	+	–	–	+	+	–	–	–
–	–	+	–	–	–	–	+	–
–	+	+	–	+	–	–	+	+
+	+	+	+	–	+	+	–	+
–	–	–	+	–	+	+	+	+
+	+	–	–	–	+	+	–	–
–	+	–	+	–	+	+	+	–
+	+	–	+	+	+	+	+	–
+	–	+	–	–	–	–	–	–
–	–	+	–	–	+	+	–	+
–	–	+	–	+	+	–	–	–
+	–	–	+	–	+	–	–	+
+	+	–	–	–	+	–	+	+
–	–	–	–	+	+	+	+	–
–	+	–	–	–	+	+	–	+
–	–	–	+	+	+	+	–	+
+	–	–	–	–	–	–	+	+
–	–	+	+	+	+	+	+	+
–	+	+	–	–	–	–	–	–
–	–	+	+	–	+	+	–	–
+	+	+	–	–	–	+	–	+

–	+	+	+	–	–	+	+	+
–	+	+	+	+	–	–	+	–
+	–	–	–	–	–	+	+	–
+	–	+	+	+	+	+	–	–
–	+	+	+	–	+	–	+	+
–	+	–	–	+	–	+	+	+
–	+	–	+	–	–	–	–	+
+	+	+	+	+	–	+	–	–
–	–	–	+	+	–	+	+	–
–	–	+	–	+	–	+	–	–
+	+	+	+	–	–	–	+	–
–	–	–	–	–	–	–	–	–
–	–	+	+	+	–	–	–	+
+	+	+	–	–	+	+	+	–
+	+	–	–	+	–	–	+	–

9.28. Suppose that you must design an experiment to investigate seven continuous factors. Running all factors at two levels is thought to be appropriate but that only the two-factor interactions involving factor *A* are of interest.

(a) How many runs are required to estimate all main effects and two factor-interactions?

1 for the intercept term, 7 for the main effects, and 6 for the two-factor interactions involving A, for a total of 14 runs.

(b) Find a minimum-run *D*-optimal design that is suitable for this problem.

A	*B*	*C*	*D*	*E*	*F*	*G*
–	–	–	+	–	–	+
+	+	–	+	–	–	–
–	+	+	–	–	–	+
–	+	–	–	+	+	+
+	+	+	+	+	+	–
+	–	+	+	–	–	+
–	+	+	+	+	–	–
+	–	–	–	+	+	+
+	+	+	–	–	+	+
+	+	+	–	+	–	–
–	+	–	–	–	–	–
+	–	+	–	–	+	–
–	–	+	–	+	+	–
–	+	+	+	–	+	–

9.29S. Suppose that you must design an experiment to investigate six continuous factors. It is thought that running all factors at two levels is adequate but that only the *AB*, *AC*, and *AD* two-factor interactions are of interest.

(a) How many runs are required to estimate all main effects and two factor-interactions?

1 for the intercept term, 6 for the main effects, and 3 for the two-factor interactions involving *AB*, *AC* and *AD*, for a total of 10 runs.

(b) Find a minimum-run *D*-optimal design that is suitable for this problem.

A	*B*	*C*	*D*	*E*	*F*

+	+	–	–	–	–
+	+	+	+	+	+
–	–	–	+	–	+
+	–	+	+	–	–
+	–	–	+	+	+
–	–	+	+	+	–
–	–	+	–	–	+
–	+	–	–	+	–
+	–	+	–	+	+
–	+	+	+	–	+

9.30. Suppose that you must design an experiment with six categorical factors. Factor *A* has six levels, factor *B* has five levels, factor *C* has five levels, factor *D* has three levels, and factors *E* and *F* have two levels. You are interested in main effects and two factor interactions.

(a) How many runs are required to estimate all of the effects that are of interest?

Intercept	1	*AB*	20	*BE*	4
A	5	*AC*	20	*BF*	4
B	4	*AD*	10	*CD*	8
C	4	*AE*	5	*CE*	4
D	2	*AF*	5	*CF*	4
E	1	*BC*	16	*DE*	2
F	1	*BD*	8	*DF*	2
				EF	1
Total	131				

(b) Find a *D*-optimal design suitable for this problem.

Order	*A*	*B*	*C*	*D*	*E*	*F*
1	L2	L5	L4	L1	L1	L1
2	L5	L3	L5	L2	L1	L2
3	L3	L2	L4	L3	L2	L1
4	L5	L4	L2	L1	L1	L2
5	L1	L2	L3	L2	L2	L2
6	L4	L1	L3	L1	L1	L1
7	L1	L3	L2	L2	L1	L2
8	L6	L5	L1	L1	L2	L1
9	L5	L4	L5	L1	L2	L2
10	L4	L2	L5	L2	L2	L1
11	L2	L2	L1	L3	L1	L1
12	L5	L1	L1	L2	L2	L2
13	L2	L3	L5	L2	L2	L2
14	L4	L5	L3	L2	L2	L2
15	L3	L4	L2	L3	L2	L2
16	L6	L5	L3	L1	L1	L2
17	L5	L2	L1	L2	L2	L2
18	L4	L1	L4	L3	L1	L1
19	L5	L3	L2	L3	L2	L2
20	L1	L4	L5	L2	L1	L1
21	L6	L4	L4	L1	L1	L1
22	L4	L2	L3	L1	L2	L2
23	L2	L3	L1	L2	L2	L1
24	L3	L5	L5	L3	L1	L2
25	L2	L1	L3	L2	L1	L2
26	L2	L1	L5	L1	L2	L1
27	L6	L2	L2	L2	L2	L2
28	L1	L4	L1	L1	L2	L2
29	L3	L5	L4	L2	L2	L1
30	L4	L3	L3	L3	L2	L2
31	L4	L1	L5	L2	L1	L2
32	L2	L3	L2	L1	L1	L1
33	L6	L2	L4	L3	L1	L2
34	L5	L5	L1	L3	L2	L1
35	L1	L5	L2	L1	L1	L1
36	L3	L2	L5	L3	L2	L2
37	L4	L4	L2	L2	L2	L1
38	L2	L5	L3	L3	L2	L2

Order	*A*	*B*	*C*	*D*	*E*	*F*
67	L2	L1	L4	L1	L2	L2
68	L6	L3	L5	L3	L2	L1
69	L3	L2	L2	L2	L1	L2
70	L5	L4	L3	L2	L2	L1
71	L4	L4	L5	L3	L1	L2
72	L1	L1	L1	L1	L1	L1
73	L4	L3	L4	L2	L1	L2
74	L6	L5	L2	L3	L2	L1
75	L3	L2	L3	L2	L1	L1
76	L5	L5	L2	L1	L2	L2
77	L6	L1	L3	L3	L2	L1
78	L2	L4	L1	L3	L1	L2
79	L6	L2	L5	L1	L2	L1
80	L1	L3	L4	L2	L2	L1
81	L1	L5	L1	L3	L1	L1
82	L2	L3	L4	L3	L2	L1
83	L3	L4	L5	L1	L1	L1
84	L4	L4	L3	L2	L1	L2
85	L2	L1	L2	L1	L1	L2
86	L3	L3	L2	L2	L2	L1
87	L4	L1	L1	L3	L1	L2
88	L5	L1	L4	L1	L2	L1
89	L2	L4	L5	L3	L2	L1
90	L6	L5	L5	L2	L1	L1
91	L4	L4	L1	L2	L2	L2
92	L2	L5	L2	L2	L1	L1
93	L1	L5	L3	L1	L2	L1
94	L5	L1	L5	L3	L1	L1
95	L3	L3	L4	L3	L1	L2
96	L6	L1	L4	L2	L2	L1
97	L1	L3	L3	L3	L1	L1
98	L1	L5	L5	L2	L2	L2
99	L6	L4	L1	L3	L2	L1
100	L4	L2	L2	L3	L1	L2
101	L2	L2	L2	L1	L2	L1
102	L3	L4	L4	L2	L1	L2
103	L6	L2	L3	L3	L1	L1
104	L4	L1	L1	L2	L2	L1

39	L1	L2	L4	L1	L1	L2
40	L6	L2	L1	L2	L1	L2
41	L3	L2	L1	L1	L1	L2
42	L5	L5	L3	L3	L1	L1
43	L1	L4	L4	L3	L2	L2
44	L5	L1	L2	L2	L1	L1
45	L3	L3	L3	L1	L2	L1
46	L6	L4	L5	L1	L1	L2
47	L2	L2	L3	L1	L1	L2
48	L4	L1	L2	L1	L2	L2
49	L5	L3	L1	L1	L1	L2
50	L2	L4	L4	L2	L1	L1
51	L4	L5	L4	L3	L1	L2
52	L6	L3	L3	L2	L1	L1
53	L3	L1	L1	L3	L1	L1
54	L1	L2	L5	L1	L2	L1
55	L2	L4	L2	L2	L2	L2
56	L5	L2	L2	L1	L1	L1
57	L4	L3	L1	L3	L1	L1
58	L6	L4	L3	L2	L2	L2
59	L1	L1	L5	L3	L2	L2
60	L3	L5	L4	L1	L2	L2
61	L4	L5	L5	L1	L1	L1
62	L3	L3	L1	L3	L2	L2
63	L2	L2	L4	L2	L1	L2
64	L3	L4	L3	L3	L1	L1
65	L6	L1	L1	L1	L2	L2
66	L5	L4	L1	L2	L1	L1

105	L6	L5	L5	L3	L2	L2
106	L4	L2	L1	L1	L1	L1
107	L2	L1	L2	L3	L2	L1
108	L3	L3	L5	L2	L1	L1
109	L4	L4	L4	L1	L2	L1
110	L5	L2	L3	L3	L1	L2
111	L3	L1	L3	L2	L2	L2
112	L2	L2	L5	L3	L1	L2
113	L6	L3	L4	L1	L2	L2
114	L1	L4	L2	L3	L1	L1
115	L3	L4	L1	L2	L2	L1
116	L5	L5	L4	L2	L2	L2
117	L2	L4	L3	L1	L2	L1
118	L1	L2	L1	L3	L2	L1
119	L4	L3	L5	L1	L2	L1
120	L6	L1	L2	L3	L1	L2
121	L5	L2	L5	L3	L2	L1
122	L5	L3	L3	L1	L2	L1
123	L1	L5	L4	L2	L1	L2
124	L6	L4	L2	L1	L2	L1
125	L2	L5	L1	L1	L2	L2
126	L3	L5	L1	L2	L1	L2
127	L4	L3	L2	L1	L1	L2
128	L1	L1	L2	L2	L2	L1
129	L5	L4	L4	L3	L1	L1
130	L2	L2	L3	L2	L2	L1
131	L3	L3	L5	L1	L1	L2

(c) Suppose that the experimenter decides that this is too many runs. What strategy would you recommend?

A design for main effects could be initially run with only 18 runs which would identify the significant factors. This could be followed by a second experiment that would include the significant factors and levels of interest.

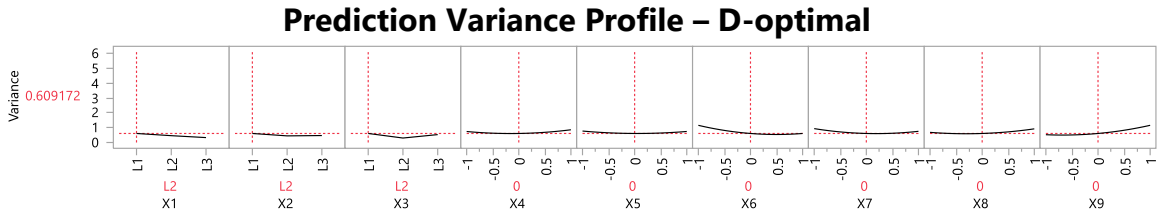
9.31. Suppose that you need to design an experiment for 9 factors. Three of these factors are categorical with three levels, and the remaining six are continuous. You are interested in the main effects and the two-factor interactions.

(a) How many runs are required for this experiment?

76 runs are required.

(b) Construct a *D*-optimal design with the minimum number of runs. What are the relative variances of the model parameters?

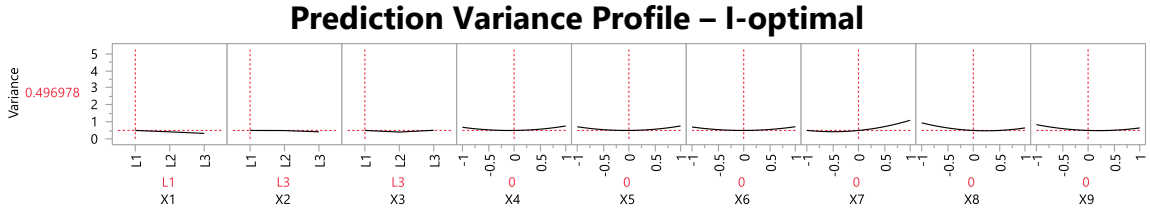
JMP Output



(c) Construct an *I*-optimal design with the minimum number of runs. What are the variances of the model parameters? How does this design compare to the *D*-optimal design you constructed in part (b)?

The *I*-optimal design has smaller variances for the factors. *I*-optimal designs minimize variance. For comparison, the *D*-efficiency of the *D*-optimal design is 73.50353, the *D*-efficiency of the *I*-optimal design is 64.37547.

JMP Output



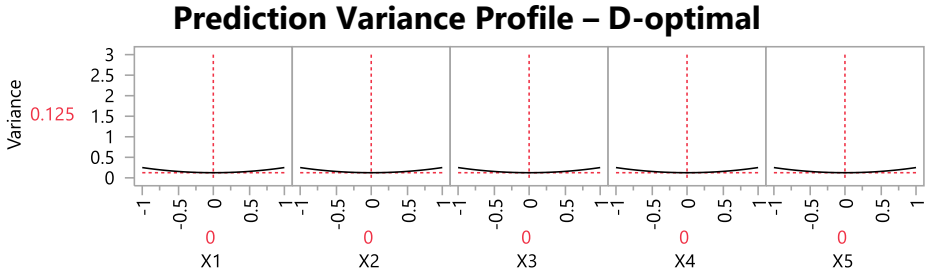
9.32S. Suppose that you need to design an experiment for 5 factors. You are interested in the main effects for all factors, but only the *AB* and *AC* factor interactions.

(a) How many runs are required for this experiment?

8 runs are required.

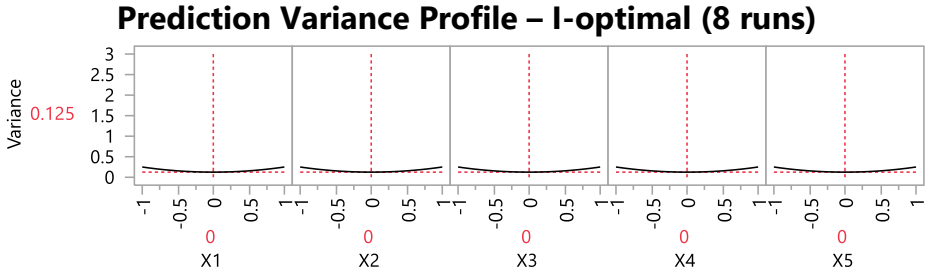
(b) Construct a *D*-optimal design with the minimum number of runs. What are the relative variances of the model parameters?

JMP Output



(c) Construct an *I*-optimal design with the minimum number of runs. What are the relative variances of the model parameters? How does this design compare to the *D*-optimal design you constructed in (b)? The designs are the same.

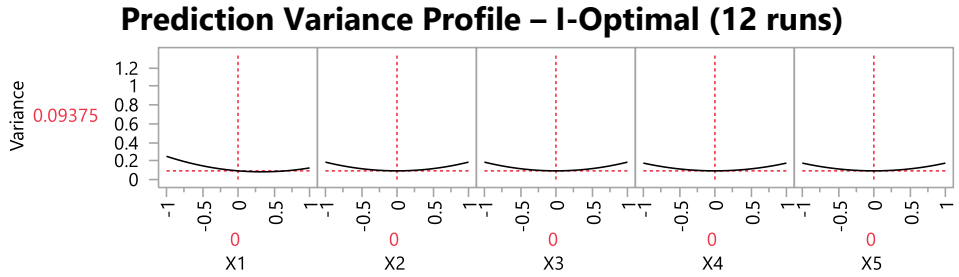
JMP Output



(d) Construct and *I*-optimal design with 12 runs. How does this design compare to the minimum-run design from part (c)?

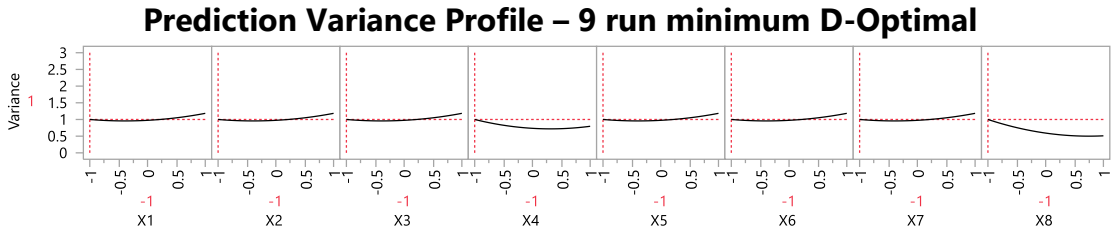
The variance has decreased from 0.125 to 0.09375. The *D*-efficiency has decreased from 1 to 95.67926.

JMP Output

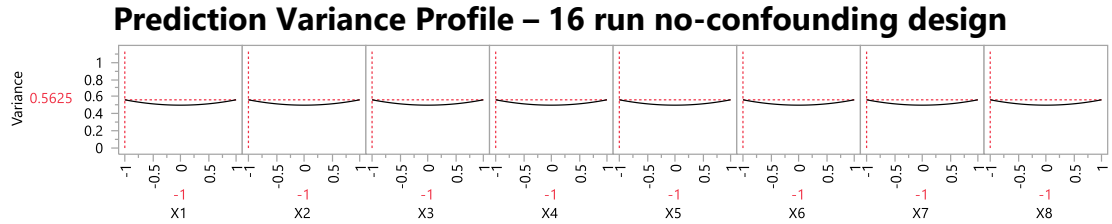


9.33. Construct a minimum run *D*-optimal design for 8 two-level factors. Design this experiment to estimate all main effects and as many two-factor interactions as possible. Compare this design to the 16-run no-confounding design for 8 factors. Which design would you prefer? Why?

JMP Output



JMP Output



The 16 run design is balanced – the variance profiles are very uniform and less than the 9 run *D*-optimal. The *D*-efficiency of the 16 run design is 1.00, the *D*-efficiency for the 9 run design is 93.19834. The nine run design is not bad if 9 runs is all you can afford.

9.34. Suppose that you need to conduct an experiment for 4 categorical factors. Factor *A* has 5 levels, factor *B* has 4 levels and the other two factors each have 3 levels. You are interested in all main effects and the two-factor interactions.

(a) How many runs are required?

56 runs are required.

(b) How many runs would you actually recommend?

60 runs are recommended.

(c) Find an appropriate *D*-optimal design.

Run	*A*	*B*	*C*	*D*
1	L3	L1	L1	L3
2	L2	L2	L2	L3
3	L1	L4	L2	L2
4	L4	L3	L2	L2
5	L5	L1	L3	L1
6	L3	L2	L1	L1
7	L2	L3	L2	L3
8	L4	L4	L2	L3
9	L1	L4	L1	L1
10	L5	L2	L1	L3
11	L4	L3	L3	L1
12	L1	L3	L2	L1
13	L1	L3	L3	L2
14	L2	L2	L1	L2
15	L3	L2	L2	L2
16	L5	L4	L1	L3
17	L4	L1	L2	L2
18	L1	L4	L3	L3
19	L3	L3	L1	L2
20	L4	L2	L3	L3

Run	*A*	*B*	*C*	*D*
21	L4	L2	L2	L1
22	L3	L4	L3	L1
23	L5	L1	L2	L3
24	L1	L4	L1	L2
25	L2	L1	L3	L3
26	L2	L3	L1	L1
27	L5	L2	L2	L2
28	L4	L4	L1	L1
29	L5	L4	L2	L1
30	L1	L3	L1	L3
31	L1	L1	L3	L1
32	L4	L2	L1	L2
33	L1	L2	L2	L3
34	L2	L1	L2	L1
35	L2	L4	L3	L1
36	L4	L3	L1	L3
37	L2	L4	L1	L3
38	L1	L2	L3	L2
39	L5	L1	L1	L2
40	L3	L4	L2	L3

Run	*A*	*B*	*C*	*D*
41	L3	L1	L2	L1
42	L5	L4	L3	L2
43	L3	L1	L3	L2
44	L4	L4	L3	L2
45	L3	L4	L1	L2
46	L1	L1	L1	L3
47	L5	L2	L3	L1
48	L5	L3	L2	L2
49	L1	L1	L2	L2
50	L3	L2	L3	L3
51	L3	L3	L2	L1
52	L2	L4	L2	L2
53	L4	L1	L1	L1
54	L5	L3	L3	L3
55	L3	L3	L3	L3
56	L2	L3	L3	L2
57	L5	L3	L1	L1
58	L2	L1	L1	L2
59	L1	L2	L1	L1
60	L4	L1	L3	L3